

Autodesk Apprenticeship

Round 1: Online Assessment

The Online Assessment consisted of four coding questions with increasing difficulty levels.

1. Question 1 (Easy)

- Coding problem of easy difficulty level.
- Exact problem statement is not available.

2. Question 2 (Easy)

- Coding problem of easy difficulty level.
- Exact problem statement is not available.

3. Question 3 (Medium)

- Given two arrays:
 - The first array represented the durability/capacity of each battery.
 - The second array represented the time required to recharge each battery.
- A system needed to run continuously for a specified duration using battery backups.
- The objective was to determine the minimum number of battery recharges required to maintain operation for the given time period.

4. Question 4 (Hard)

- Given an array of integers, one operation allowed incrementing or decrementing an element by 1 at a cost of 1.
- The task was to transform the entire array into either:
 - a strictly increasing sequence, or
 - a strictly decreasing sequence,
- while minimizing the total modification cost.

Round 2: HireVue Video Interview

The second stage of the recruitment process was a HireVue interview consisting of seven questions. The interview evaluated both technical knowledge and behavioral competencies.

The assessment included:

- 6 video-response questions, each with a response time of approximately 2–3 minutes.
- 1 written-response question requiring an answer of approximately 50 words.

Since the role was Software Development Engineer (SDE) focused, the majority of the questions assessed technical background, programming knowledge, and communication skills.

Questions Asked:

1. Self-Introduction

- Introduce yourself and provide an overview of your academic and technical background.

2. Programming Languages

- Name three programming languages you are proficient in, rank them based on your expertise, and explain the scenarios in which you would use each language.

3. Role Suitability

- Explain why you believe you are a strong candidate for the Software Development Engineer role.

4. Communication Skills

- Describe a situation where you effectively demonstrated communication and collaboration skills.

5. Technical Stack

- Explain the technologies, programming languages, frameworks, and tools that you use most frequently.

6. Additional Information

- Share any additional information, achievements, experiences, or qualities that you would like the company to know about you.

Written Response Question:

7. Favorite Programming Language

- Describe your favorite programming language and explain the reasons for your preference in approximately 50 words.

This round primarily assessed technical expertise, communication abilities, problem-solving mindset, role alignment, and overall professional suitability for the position.

Round 3: Technical Interview

The first technical interview focused on understanding the candidate's background, project experience, problem-solving abilities, software engineering mindset, and fundamental computer science concepts.

Discussion Topics and Questions:

1. Introduction

- Brief introduction covering academic background, technical skills, projects, and areas of interest.

2. Career Aspirations

- Discussion regarding future plans, including whether the candidate intended to pursue higher education or build a career in the software industry.

3. Project Discussion

- In-depth discussion of projects mentioned in the resume.
- Questions focused on project architecture, technologies used, design decisions, implementation details, challenges faced, and individual contributions.

4. Data Structures and Algorithms Problem

- Given two arrays, **Primary** and **Secondary**.
- Two types of queries were supported:
 - **Update Query:** Replace the element at a specified index in the Secondary array with a new value.
 - **Count Query:** Given a target value, determine the number of pairs (i, j) such that:
$$\text{Primary}[i] + \text{Secondary}[j] = \text{Target}$$

5. Software Engineering Scenario

- A situation was presented where the candidate was assigned a problem statement independently, without direct assistance from team members.
- The interviewer asked how the candidate would approach understanding requirements, researching solutions, utilizing

available resources (including AI tools), implementing the solution, testing it, and validating the results.

6. Operating Systems Fundamentals

- Discussion on the candidate's understanding of Operating Systems.

7. Programming Language Preference

- Discussion on the candidate's preferred programming language and the reasons for choosing it, including its strengths, use cases, and personal experience with the language.

This round primarily assessed technical fundamentals, project knowledge, problem-solving skills, software engineering approach, and communication abilities.

Round 4: Managerial Interview

The managerial round focused on evaluating the candidate's project experience, technical decision-making ability, problem-solving approach, teamwork skills, and overall suitability for working in a collaborative engineering environment.

Discussion Topics and Questions:

1. Introduction

- Brief discussion covering academic background, technical interests, projects, and career aspirations.

2. Project Discussion and Technical Decision-Making

- Detailed discussion of projects undertaken by the candidate.
- Questions focused on:
 - Project architecture and design choices.
 - Performance optimizations implemented in the project.
 - Trade-offs considered while selecting technologies, frameworks, and tools.
 - Evaluation of alternative approaches and the rationale behind final decisions.
 - Scalability, maintainability, and efficiency considerations.

3. Scenario-Based Data Structures and Algorithms Discussion

- Multiple scenario-based questions related to data structures and problem-solving.
- Approximately five scenarios were discussed, each followed by several follow-up questions.
- The interviewer focused on:
 - Selection of appropriate data structures.
 - Time and space complexity trade-offs.
 - Optimization strategies.
 - Real-world application of algorithmic concepts.
 - Ability to justify technical decisions under varying constraints.

4. Programming Language Preference

- Discussion on the reasons for preferring C++.
- Topics included performance characteristics, memory management capabilities, Standard Template Library (STL), object-oriented programming support, and its suitability for systems programming and algorithmic problem-solving.

5. Team Collaboration Experience

- The candidate was asked to describe a situation where they worked effectively within a team.
- The discussion focused on collaboration, communication, responsibility sharing, conflict resolution, and contribution toward achieving a common objective.

6. Effective Teamwork

- Discussion on the most important factors required for successful teamwork.
- Topics included communication, trust, accountability, collaboration, adaptability, and alignment toward shared goals.

This round primarily assessed technical decision-making, leadership potential, collaboration skills, problem-solving ability, communication effectiveness, and overall readiness to work within a professional software engineering team.